

WASP-43b

R-2594

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140218 · created 2026-08-27T23:44:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456706.7801 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.014
Transit depth (Rp/Rs)^2	2.92 +/- 0.47 [%]
Orbital Inclination (inc)	82.44 +/- 1.51
Airmass coefficient 1 (a1)	0.974 +/- 0.012
Airmass coefficient 2 (a2)	0.018 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#3 - [101, 216]
Optimal Aperture	2.11
Optimal Annulus	10.68
Transit Duration (day)	0.0523 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.014 vs 0.1594 (deviation 0.83σ)
Duration fit vs published	0.0523 d vs 0.0512 d (deviation 0.22σ)
Mid-time O–C	1.3 min
Depth SNR	12.2
Residual scatter	1.2 %

Light curve

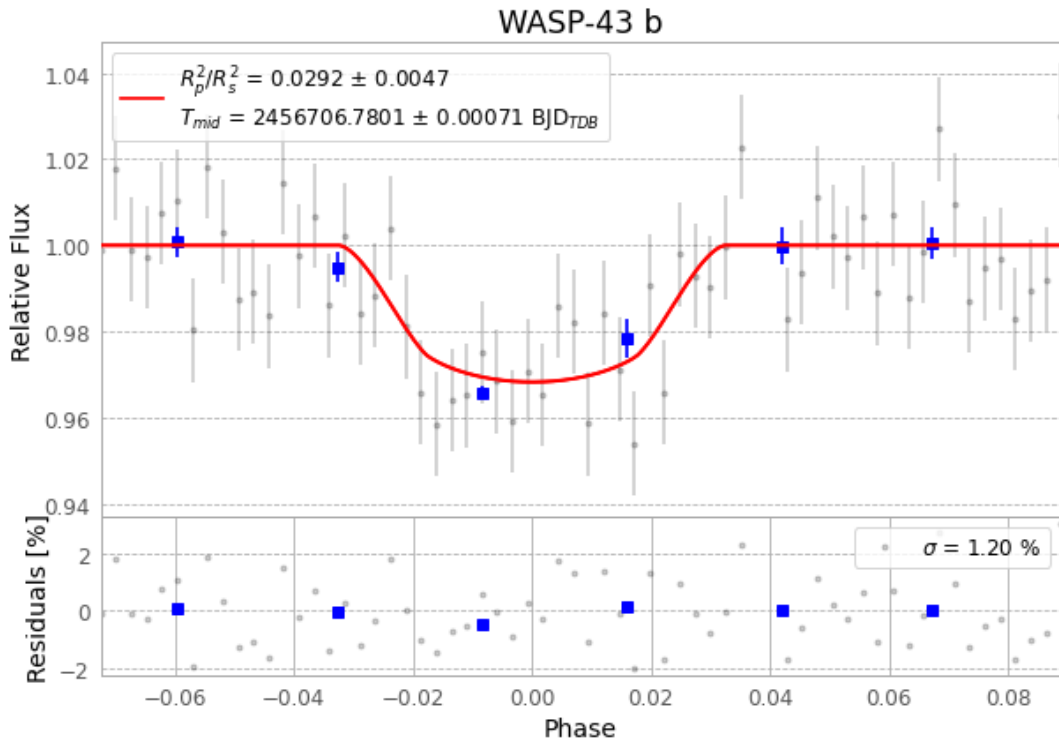


Figure 1 — FinalLightCurve_WASP-43 b_2014-02-17.png (SHA-256 c0ac4656034b588f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [167, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 511a8e00a46815b3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140218050912.FITS → WASP-43140218081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140218.log (4c41d82f3764...), FinalLightCurve_WASP-43 b_2014-02-17.png (c0ac4656034b...), FinalLightCurve_WASP-43 b_2014-02-17.csv (d05331272d57...)

Compute resources

Wall time 107.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:44Z.